

Part 15: Subroutine with Parameters

1. **How do you pass parameters through registers?** Simply `MOV` data into registers (like `AX`, `BX`) before the `CALL`. The subroutine reads those registers.
2. **How do you pass parameters through stack?** `PUSH` the values onto the stack before the `CALL`. Inside the subroutine, access them using `BP` relative to the stack frame.
3. **How do you access parameters inside subroutine?** If passed via stack: `MOV BP, SP` and access like `[BP+4]`. If via registers: just use the register names.
4. **How do you return values from subroutine?** Typically, place the result in the `AX` register before executing `RET`.
5. **Which registers are commonly used for parameters?** `AX`, `BX`, `CX`, and `DX` are common.
6. **How do you pass arrays as parameters?** Pass the **address** (offset) of the array in a register like `SI` or `BX` (`LEA SI, MyArray`).
7. **How do you pass byte vs word parameters?** Just ensure you use the correct register size (e.g., `AL` for byte, `AX` for word) or `PUSH` the correct size.
8. **How do you increment pointer to access multiple parameters?** If passing an array address in `SI`, use `INC SI` (for bytes) or `ADD SI, 2` (for words) inside the loop.
9. **How do you implement subroutine with conditional execution?** Pass a flag or value. Inside the subroutine, use `CMP` and jumps to decide which parts to run.
10. **How do you handle nested subroutine calls with parameters?** Each subroutine should protect its own registers (`PUSH/POP`) so inner calls don't overwrite the outer caller's data.
11. **How do you preserve registers while using parameters?** `PUSH` any registers you intend to change at the start of the procedure and `POP` them at the end.
12. **How do you pass constants as parameters?** Move the immediate value into a register (`MOV AX, 5`) before calling.
13. **How do you pass memory addresses to subroutine?** Use `LEA` (Load Effective Address) to put the variable's offset into a register (e.g., `LEA BX, var`).
14. **How do you combine multiple subroutines with parameters?** Break a complex task into small subroutines (e.g., `Calculate`, then `Print`), passing the output of one as input to the next.
15. **How do you debug parameter passing in EMU8086?** Use "Single Step" and watch the **Registers** window and the **Stack** memory view to see values being pushed and popped.